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Magnesium salt promoted tandem nucleophilic addition–Oppenauer oxidation of aldehydes with organozinc reagents†

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A magnesium salt promoted synthesis of ketones via tandem nucleophilic addition–Oppenauer oxidation of aldehydes using organozinc chemistry was demonstrated. Magnesium salts concomitantly generated via magnesium metal mediated organohalide zincation exhibit high efficacy for nucleophilic addition of organozinc reagents to aromatic aldehydes and thereafter Oppenauer oxidation whereby ketones are formed in high to excellent yields.

A tandem reaction is widely recognized as a promising protocol for C–C bond formation, functional group implantation and transformation in a one-pot operation.¹ In this respect, strategies involving reactive organometallics are still rare.² The nucleophilic addition/Oppenauer oxidation of aldehydes³ has proven to be a practical and straightforward protocol for ketone synthesis. Among them, the Mg–Oppenauer protocol^{3g} provided by Knochel *et al.* was considered to be the most advantageous as the *in situ* formed magnesium alkoxides are intrinsically powerful promoters for the forthcoming hydride transference, thus allowing THF, the most commonly used solvent for Grignard reagents albeit a poor solvent for Oppenauer oxidation,^{3b–d} which can be directly used without apparent inhibition to hydride transference. Besides, the environmentally friendly benzaldehyde, instead of carcinogenic chromium salts, enormous reagent excess (MnO₂) or explosively exothermic reagents (oxalyl chlorides/DMSO), was employed as an oxidant. However, as we⁴ previously mentioned, Grignard reagents should be accurately titrated beforehand and transferred carefully in case over- or less-loading will lead to significant side reactions such as Aldol condensation and Tishchenko esterifications.⁵ Additionally, routes starting from Grignard reagents demand low reaction temperatures, and sensitive functional groups are still found difficult to tolerate.⁶ To circumvent these problems, we believe that reactive

Grignard reagents should be transmetalated to a weaker organometallic reagent or, more directly, a milder organometallic reagent should be used. In this respect, organozinc reagents arise as an ideal candidate owing to their mild reactivity and tolerance for a wide range of functional groups.⁷ Based on our interest in organozinc reagents,⁸ we were interested to employ organozinc reagents as nucleophiles in the nucleophilic addition/Oppenauer oxidation of aldehydes whereby highly functionalized ketones could be expediently produced.

At the beginning, pivaldehyde⁹ was selected as the hydride acceptor owing to its low dielectric constant and high oxidation potential ($E_0 = 211$ mV).¹⁰ Thus, the reaction of Et₂Zn **1a** and 4-methoxybenzaldehyde **2a** was first examined to evaluate the efficiency of organozinc reagents (Table 1). Undoubtedly, this reaction did not occur without any additive¹¹ (entry 1). Metallic salts such as LiX,¹² MgCl₂,¹³ CoBr₂,¹⁴

Table 1 Optimization of reaction conditions^a

Entry	Additive ^b	3a ^c (%)	3a/4a ^c
1		n. r. ^e	—
2	LiCl	n. r.	—
3	MgCl ₂	n. r.	—
4	MgBr ₂	92 (84) ^d	20 : 1
5	MgI ₂	90 (82) ^d	15 : 1
6	MgCl ₂ ·LiCl	57	2 : 1
7	MgBr ₂ ·LiCl	90 (82) ^d	18 : 1

^a The reaction was performed with **1a** (4.0 mmol), **2a** (2.0 mmol), and pivaldehyde (4.0 mmol) in THF (10 mL) under 0 °C and then stirred at room temperature. ^b Additive (4.0 mmol). ^c Yields and the product ratios were determined by crude ¹H NMR. ^d Isolated yield of **3a**. ^e n. r. = no reaction.

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NiCl_2 ¹⁵ *etc.* have a significant positive effect on both the formation of organometallic reagents and their reactivity. LiCl , for example, has been extensively explored by Knochel *et al.* as a key additive for the formation of organometallic reagents, such as Grignard reagents,⁶ and organozinc,¹⁶ organoindium¹⁷ and organoaluminum¹⁸ reagents. In our reaction system, however, LiCl alone was tested as an inert additive and no reactions were detected at all (entry 2). MgCl_2 has been frequently used as a key promoter in a plethora of organozinc reagents involving reactions.¹⁹ Unfortunately, it is still an invalid additive here, probably because of its poor solubility in THF.²⁰ Gratifyingly, one equivalent addition of MgBr_2 promptly initiated and completed this transformation within 2 hours, giving ketone **3a** in nearly quantitative yield and excellent chemoselectivity (**3a** : **4a** = 20 : 1) (entry 4). MgI_2 too, is a powerful additive, delivering **3a** in 90% yield (82% isolated yield, entry 5). Combination of an equal amount of LiCl and MgCl_2 made a clear solution in THF, allowing the addition reaction to be completed in 6 h, albeit the desired ketone **3a** was obtained in only 57% isolated yield together with a significant amount of secondary alcohol **4a** that remained (entry 6). An equal molar mixture of LiCl and MgBr_2 also cleanly converted aldehyde **2a** into ketone **3a** in a similar yield as with MgBr_2 alone (entry 7). These experimental facts implied that LiCl did not perform any pivotal role besides increasing the solubility of magnesium salts in THF (entries 2–7). Other metallic salts screened such as CsF , CoCl_2 , NiCl_2 , CH_3OLi , and $\text{Al}(\text{OiPr})_3$ all failed in promoting this type of transformation.

With these optimized conditions in hand, we first examined the applicability of organozinc reagents. As summarized in Table 2, a wide range of aromatic or heteroaromatic aldehydes could be successfully converted into the corresponding ketones using diorganozinc reagents such as Me_2Zn (**3b–3d**, **3p**), Et_2Zn (**3e–3g**) and Ph_2Zn (**3h–3n**). Neither electron donating nor electron accepting substituents appear to influence the reaction course. Various heteroaromatic ketones, such as thiophenyl (**3g**, **3m**), furyl (**3l**) and pyridyl (**3n**) ketones are easily prepared in good yields from the corresponding aldehydes. Notably, in the presence of 2 equiv. of pivaldehydes, zinc, LiCl and MgBr_2 , the Reformatsky type reaction of 4-methoxybenzaldehyde with ethyl 2-bromopropionate gave the desired ketone **3o** in good yield. Importantly, ferrocenyl ketone **3p** was obtained from ferrocenecarbaldehyde in 72% yield.

Although diorganozinc reagents are effective organometallic species here, highly functionalized diorganozinc reagents are not easily available.²¹ It is advantageous to employ functionalized organozinc halides as they can be easily prepared from cheap bromides by ZnCl_2 mediated magnesium insertion.²² Most interestingly, magnesium salts concomitantly generated during organozinc halide formation offer an immediate promoter for this type of transformation. Thus, treatment of $\text{PhZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl}$ **5a**, prepared by *in situ* magnesiation of phenyl bromide in the presence of ZnCl_2 and LiCl , with a mixture of piperonyl aldehyde and two equivalents of pivaldehyde at room temperature, yielded ketone **6a** in 79% isolated yield (Table 3). Notably, only less than 7% byproducts,

Table 2 Reaction of aldehydes with organozinc reagents^a

$\text{R}-\text{CHO} \xrightarrow[\text{THF, 0}^\circ\text{C to RT}]{\text{R}'_2\text{Zn, MgBr}_2, \text{LiCl (1) pivaldehyde (2.0 equiv)}} \text{R}-\text{CO}-\text{R}'$	
	3b , X = OMe, 72% 3c , X = Cl, 77% 3d , X = Br, 74% 3e 80% 3f 78% 3g 63% 3h , X = OMe, 82% 3i , X = NMe ₂ , 71% 3j 62% 3k 72% 3l , X = O, 54% 3m , X = S, 75% 3n 72% 3o 76% ^b 3p 72%

^a The reaction was performed with $\text{R}'_2\text{Zn}$ **1** (4.0 mmol), aldehyde **2** (2.0 mmol), pivaldehyde (4.0 mmol), MgBr_2 (4.0 mmol) and LiCl (4.0 mmol) in THF (10 mL) under an ice-water bath and then stirred at room temperature. ^b Zn (4.0 mmol) and $\text{BrCH}_2\text{CO}_2\text{Et}$ (4.0 mmol) were used instead of $\text{BrZnCH}_2\text{CO}_2\text{Et}$.

Table 3 Reaction of aldehydes with organozinc halides^a

$\text{Ar}-\text{CHO} \xrightarrow[\text{THF, 0}^\circ\text{C to RT, 2h}]{\text{RZnBr}\cdot\text{MgCl}_2\cdot\text{LiCl } \mathbf{5}} \text{Ar}-\text{CO}-\text{R}$	
	6b , R = 2-Me, 83% 6c , R = 2-OMe, 76% 6d , R = 4-F, 80% 6e , R = 4-Cl, 81% 6f , R = 2-Cl, 75% 6g , R = 3-NO ₂ , 77% 6h , R = 4-CN, 75% 6i , R = 4-CO ₂ Me, 70% 6a 79% 6j 71% 6k 58% 6l 65% 6m 56% 6n 82% 6o 74%

^a Organozinc reagents were added to a mixture of aldehyde (2.0 mmol) and pivaldehyde (4.0 mmol) under an ice-water bath and were then stirred at room temperature for 2 h.

mainly derived from nucleophilic addition–Oppenauer oxidation of the organozinc reagent **5a** to pivaldehyde, were detected according to GC–MS analysis. This promising result encouraged us to further examine the efficacy of other aromatic aldehydes. Gratifyingly, aromatic aldehydes bearing both electron-donating groups such as methyl (**6b**) and methoxyl (**6c**) and electron-withdrawing groups, such as fluoro (**6d**), chloro (**6e**, **6f**), nitro (**6g**), nitrile (**6h**) and ester (**6i**) produced the desired ketones with good to excellent results. Steric hindered aromatic aldehydes, such as 2-methyl, 2-methoxyl and 2-chlorobenzaldehydes are applicable with the corresponding ketones (**6b**, **6c**, **6f**) obtained in satisfactory yields. Heteroaromatic organozinc reagents as represented by 2-furanyl, 2- and 3-thiophenylzinc halides were well applied, giving the corresponding heteroaromatic ketones (**6k–6m**) in acceptable yields. Sensitive functional groups such as nitro (**6g**), nitrile (**6h**) and ester (**6i**) were well tolerated with the corresponding functionalized ketones obtained in good yields. Again, ferrocenyl ketones **6n** and **6o** were efficiently prepared from ferrocene-carbaldehyde in high yields.

LiCl mediated magnesium–halogen exchange reactions are now widely accepted as a common method for the preparation of highly functionalized Grignard reagents,²³ which, in the presence of ZnCl₂, converted into the corresponding organozinc reagents.²¹ In order to examine the efficiency of the highly functionalized organozinc reagents thus prepared, ethyl 4-iodobenzoate **7a** was first magnesiated *via* I/Mg exchange with *i*-PrMgCl·LiCl²⁴ and was then treated with ZnCl₂ to give the desired organozinc reagents which exhibited excellent reactivity that the highly functionalized diaryl ketones **9a–9c** were obtained in high yields (Table 4). Iodobenzene bearing an electron-withdrawing CF₃ substituent found difficulty in direct magnesium insertion.²⁵ Nevertheless, treatment of 3-iodotrifluorobenzene by using turbo Grignard reagents (*i*-PrMgCl·LiCl) and then transmetalated with ZnCl₂ afforded the corresponding organozinc reagents, which reacted with methyl 4-formylbenzoate and 3,4,5-trimethoxybenzaldehyde in the presence of 2 equivalents of pivaldehyde, giving functionalized diaryl ketones **9d** and **9e** in 80% and 84% yields respectively (Table 4). Similarly, organozinc reagents derived from 4-fluoroiodobenzene **7c** and 4-iodobenzonitrile **7d** reacted with 4-fluorobenzaldehyde and 3,4,5-trimethoxybenzaldehyde respectively to give functionalized benzophenones **9f** and **9g** in good yields.

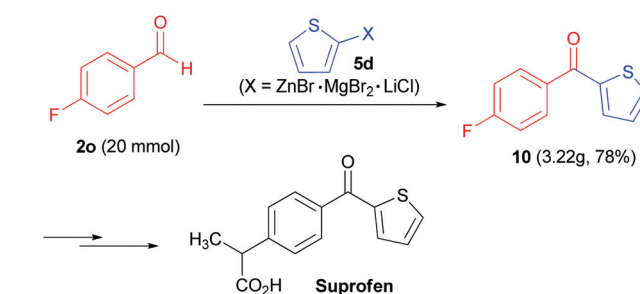
The excellent efficiency of the present reaction system was further highlighted by a 20 mmol scale synthesis of 2-(4-fluorobenzoyl)thiophene **10**, a key intermediate of a well-known medicine suprofen²⁵ (Scheme 1). Thus 2-thienylzinc bromide **5d**, readily prepared from 2-bromothiophene (30 mmol) *via* LiCl promoted zincation, underwent complete conversion of 4-fluorobenzaldehyde (20 mmol) within 2 h to afford the 2-(4-fluorobenzoyl)thiophene in 78% isolated yield.

Mechanistically, a magnesium salt plays a pivotal role both in the addition reaction of organozinc reagents towards aromatic aldehydes and Oppenauer oxidation. So we propose that the reaction proceeds as shown in Scheme 2. Initially, mag-

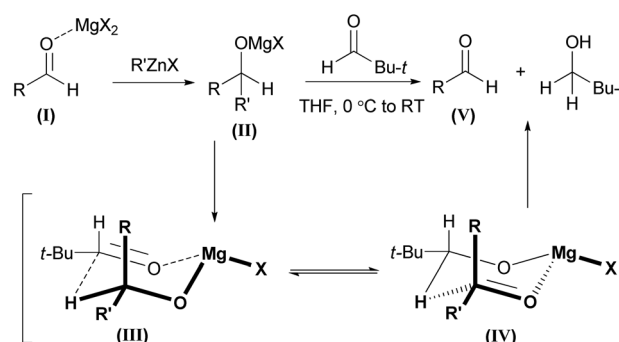
Table 4 Reaction of organozinc reagents prepared by a ZnCl₂ mediated I/Mg exchange reaction^a

(FG-R: 7a : 4-CO ₂ Et; 7b : 3-CF ₃ ; 7c : 4-F; 7d : 4-CN)	
9a , R = 4-OMe, 70%	9d , R = 4-CO ₂ Me, 80%
9b , R = 3,4-(OMe) ₂ , 74%	9e , R = 3,4,5-(OMe) ₃ , 84%
9c , R = 4-CO ₂ Me, 75%	
9f 78%	9g 72%

^a Organozinc reagents generated from I/Mg exchange (about 4 mmol) were added to a mixture of aldehyde (2.0 mmol) and pivaldehyde (4.0 mmol) under an ice-water bath and were then stirred at room temperature for 2 h.



Scheme 1 Gram scale synthesis of 2-(4-fluorobenzoyl)thiophene.



Scheme 2 Proposed reaction mechanism.

nesium salts act as a Lewis acid to complex and activate aldehyde carbonyl (**I**), which further reacts with the organozinc reagent to give a magnesium alkoxide (**II**). Pivaldehyde then

coordinates to the magnesium complex (**III**) and a hydride transfer occurs through complex (**IV**) to generate ketone (**V**).

In conclusion, a magnesium salt promoted synthesis of aromatic ketones *via* a tandem nucleophilic addition–Oppenauer oxidation of aldehydes using organozinc chemistry was demonstrated. Magnesium salts concomitantly generated *via* magnesium mediated organohalide zincation exhibit high efficacy for nucleophilic addition of organozinc reagents to aldehydes and thereafter Oppenauer oxidation whereby ketones are formed in high to excellent yields. This transformation offers a green and facile procedure for the preparation of highly functionalized aryl and alkyl ketones. The usefulness of this practical and efficient method was demonstrated in gram-scale synthesis of 2-(4-fluorobenzoyl)thiophene, a key intermediate to non-steroidal anti-inflammatory drug suprofen.

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